

PSP - Peak Planning 2022 - 2023

- Goal: [Stable Peak 2022 - 2023](#)
 - High-Level Rundown
 - Note: Data is collected from cloudwatch, for the AVG range between (10/05/2022 to 10/18/2022)

Goal: Stable Peak 2022 - 2023

High-Level Rundown

- ☐ Undo Retention Standardization
- ☐ IOPS Optimization
- ☐ Instance Type Optimization
- ☐ Redo Log Size Optimization
- ☐ Revisit allocation storage to RDS and space on undo and temp tablespace
- ☐ Revisit Processes and check other parameters
- ☐ GG Restacking for EFE
- ☐ Space allocation on Tablespace for EFE
- ☐ DR Readiness

		Details
Disk Space	AWS Console	RDS Instance configuration
Instance Class	AWS Console	RDS Instance configuration
IOPS	AWS Console	RDS Instance configuration
Number of sessions /processes	AWS Console (Monitoring)	RDS Instance Monitoring
Temp Tablespaces	SQL Query	Temp Tablespace <pre>select a.tablespace_name,a.file_name,a.bytes/1024/1024 SIZE_IN_MB,a. autoextensible,a.maxbytes/1024/1024 MAXSIZE_IN_MB, to_char(b.CREATION_TIME,'DD-MON-YYYY HH24:MI:SS') DATE_OF_BIRTH from dba_temp_files a,v\$tempfile b where a.file_name=b.name order by to_char(b.CREATION_TIME,'YYYY'),a.file_name; SELECT A.tablespace_name tablespace, D.mb_total, SUM (A.used_blocks * D.block_size) / 1024 / 1024 mb_used, D.mb_total - SUM (A.used_blocks * D.block_size) / 1024 / 1024 mb_free FROM v\$sort_segment A, (SELECT B.name, C.block_size, SUM (C.bytes) / 1024 / 1024 mb_total FROM v\$tablespace B, v\$tempfile C WHERE B.ts# = C.ts# GROUP BY B.name, C.block_size) D WHERE A.tablespace_name = D.name GROUP by A.tablespace_name, D.mb_total;</pre>
Undo retention	SQL Query	show parameters undo_retention;

Undo tablespace	SQL Query	Undo tablespace <pre> SELECT a.tablespace_name, SIZEMB, USAGEMB, (SIZEMB - USAGEMB) FREEMB FROM (SELECT SUM (bytes) / 1024 / 1024 SIZEMB, b.tablespace_name FROM dba_data_files a, dba_tablespaces b WHERE a.tablespace_name = b.tablespace_name AND b.contents like 'UNDO' GROUP BY b.tablespace_name) a, (SELECT c.tablespace_name, SUM (bytes) / 1024 / 1024 USAGEMB FROM DBA_UNDO_EXTENTS c WHERE status <> 'EXPIRED' GROUP BY c.tablespace_name) b WHERE a.tablespace_name = b.tablespace_name; select TABLESPACE_NAME, FILE_NAME, BYTES/1024/1024/1024 SIZE_IN_GB , AUTOEXTENSIBLE,MAXBYTES/1024/1024/1024 MAXGB from dba_Data_files where TABLESPACE_NAME like 'UNDO_T1'; </pre>
log group size and log groups	SQL Query	Log Groups <pre> SELECT a.group#, substr(b.member,1,30) name, a.members, a.bytes/1024/1024 bytesmb, a.status FROM v\$log a, v\$logfile b WHERE a.group# = b.group#; </pre>
DR Readiness	Jenkins Jobs	Disaster Recovery Runbook
SQL Performance		

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Database	Change	Existing Value (As of 20 Oct 2022)	Suggested Value (20 Oct 2022)	Rationale	Change ticket	Rolled back (Post Peak)	Comments
PSP Monolith	Disk Space Size	43 TB (5.8 TB free)					
	Instance Class	db.r5.16xlarge	No Change				CPU - 17.2 % Max 13 Oct 2022 12:10 PM
	IOPS	45000					9.142 K 08 Oct 2022 2:42 AM
	Latency	Read Latency Write Latency					
	log group size	1 GB	No change				WriteIOPS range- 7k - 10k ReadIOPS range - 15k - 25 k
	log groups	10	No change				
	Max Number of processes	6000	No change				
	Temp space (TEMP)	500 GB (AUTO EXTENSIBLE - 600 GB)	No change				
	Undo retention	43200	No change				

	Undo tablespace (UNDO_T1)	89 GB (AUTO EXTENSIBLE - 100GB)	No change				GB
	Add Yearly Partitions		Preprod - Pending Prod - Pending				
PSP Audit	Instance Class	db.r6g.16xlarge	No Change				CPU 6.3 % Max
	DB Connections	755	No Change				
	Maximum used Transaction IDs(count)	148 M					
	Add Yearly Partitions		Preprod - Pending Prod - Pending				